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**Department of Electronics and Communication Engineering --
KLVZA
Course Handout
2025-2026, Even Sem**

Course Title	: ELECTROMAGNETIC WAVES AND TRANSMISSION LINES
Course Code	: 24EC2209
L-T-P-S Structure	: 3-1-0-0
Pre-requisite	:
Credits	: 4
Course Coordinator	: Zia UR Rahman Mohammad
Team of Instructors	:
Teaching Associates	:
Syllabus :	CO1: Static Electric Fields: Coordinate systems, Charge distributions, Coulomb's Law & Electric field intensity, electric flux density, Gauss's Law & applications Divergence, Divergence theorem, Potential and Potential difference, Potential gradient. Boundary conditions on E and D. Electric current, current densities, equation of continuity. CO2: Static Magnetic Fields: Biot - Savart's Law and applications, Ampere's circuital law, differential form of Ampere's circuital law, Curl, Stoke's theorem, Lorentz force equation, force on a current element in magnetic field, Boundary conditions on B and H. CO3: Electromagnetic Waves: Faraday's law of Induction, Maxwell's Equations in Differential and integral forms. Wave equation for free space, Uniform plane wave-general solution and propagation. Wave equations for conducting medium. Wave equations in phasor form, wave propagation in loss less medium, Poynting vector. CO4: Transmission lines: General Solution, input impedance, infinite line, wavelength, velocity, Reflection Coefficient, Open and short circuited lines, lossless line, standing wave ratio, input impedance, open and short-circuited lossless lines, Impedance matching.
Text Books :	1. W H.Hayt & J A Buck : "Engineering Electromagnetics" TATA McGraw-Hill, 7th Edition 2007. 2. EC.Jordan, "EM waves and Radiating Systems", International Edition, 2011 3. John D Ryder , "Network Lines and fields", 2nd Edition, PHI. 4. Mathew no Sadiku, "Elements of Electromagnetics ", Oxford University Press, 2003. 5. GSN Raju, Electromagnetic Waves and Transmission Lines, Pearson Publication, 2020
Reference Books :	1. David K.Cheng: "Field and Wave Electromagnetics - Second Edition-Pearson Edition, 2004. 2. Joseph A Ed minister, "Theory and problems of Electromagnetics", 2nd edition, Scham's Outline series, Mc-Graw Hill International. 3. Constantine A. Balanis," Advanced Engineering Electromagnetics" John Wiley.
MOOCS :	https://www.coursera.org/specializations/electrodynamics#courses
Course Rationale :	Today's Electronics and Communication Engineers need to understand the fundamental principles and laws of electromagnetism to develop and implement better electronic communication system designs that build upon electromagnetic propagation and radiation effects. This can be accomplished by introducing the electromagnetic concepts, which includes basic principles of electrostatics and magnetostatics necessary for understanding the propagation characteristics of electromagnetic wave through different media. The principles of propagation of energy through transmission lines and

	waveguides have also been included with an appropriate discussion on the fundamentals of Maxwell's equations.
Course Objectives :	The course focuses on bringing thorough understanding on the essentials of electromagnetic wave propagation and its behavior while propagating through various media. The learning culminates in analyzing the characteristics of the waves through free space, propagation of energy through transmission lines, waveguides with optimized matching techniques and analyze various parameters at different frequencies. The syllabus of this course enables to apply, illustrate and analyze the various parameters and different modes of EM wave Propagation.

COURSE OUTCOMES (COs):

CO NO	Course Outcome (CO)	PO/PSO	Blooms Taxonomy Level (BTL)
CO1	Apply and choose the properties of Static Electric charges	PO1,PSO1	3
CO2	Apply and interpret the properties of Static Magnetic charges.	PO1,PSO1	3
CO3	Apply and solve various of time-varying electric and magnetic field phenomena	PO1,PSO1	3
CO4	Analyze the propagation phenomenon of EM waves in transmission lines.	PO2,PSO2	4

COURSE OUTCOME INDICATORS (COIs)::

Outcome No.	Highest BTL	COI-1	COI-2	COI-3	COI-4
CO1	3	Btl-1 Remember various vector calculus conversions over coordinate systems	Btl-2 Understand the distribution of electric charge in line, space and volume	Btl-3 Apply electrostatic laws and determine boundary conditions in time invariant electric fields	
CO2	3	Btl-1 Remember the fundamentals of magnetostatics	Btl-2 Understand various laws of magnetostatics	Btl-3 Apply magnetostatics and determine the magnetic boundary conditions	
CO3	3	Btl-1 Remember the fundamentals of Maxwells equations	Btl-2 Understand wave propagation in free space	Btl-3 Apply Maxwells equations in wave propagation	
CO4	4	Btl-1 Remembering the transmission line parameters	Btl-2 Understand various transmission line parameters	Btl-3 Apply impedance matching techniques	Btl-4 Analyze propagation characteristics of transmission lines

PROGRAM OUTCOMES & PROGRAM SPECIFIC OUTCOMES (POs/PSOs)

Po No.	Program Outcome
PO1	Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PSO1	To apply concepts of micro and nanoelectronics, embedded systems, and circuit analysis to develop smart electronic systems for real-world applications.
PSO2	To design, optimize and realize intelligent systems using next-gen technologies of signal processing, 5G/6G communications, IoT, AI, robotics and automation.

Lecture Course DELIVERY Plan:

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
1	CO1	COI-1	Introduction to Course, Course handout description, discussion on syllabus.	Course Handout	PPT	End Semester Exam,SEM-EXAM1
2	CO1	COI-1	Introduction to Static Electric Fields, Cartesian, Cylindrical & Spherical Coordinate Systems	TB-1[1] [1-3]	PPT	End Semester Exam,SEM-EXAM1
3	CO1	COI-1	Charge Distributions & Coulomb's Law	TB-1[4] [111-117], TB-1[4] [104-107]	Chalk,Talk	End Semester Exam,SEM-EXAM1
4	CO1	COI-2	Electric Field Intensity & Electric Flux Density	TB-1[4] [122-130]	PPT	End Semester Exam,SEM-EXAM1
5	CO1	COI-2	Gauss's Law and Applications	TB-1[4] [122-130]	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
6	CO1	COI-2	Divergence Theorem, Electric Potential & Potential Gradient	TB-1[3] [69-73],TB-1[4] [133-135]	PPT	End Semester Exam,SEM-EXAM1,Tutorial
7	CO1	COI-3	Boundary Conditions on E and D	TB-1[5][182-187]	Chalk,Talk	End Semester Exam,SEM-EXAM1
8	CO1	COI-3	Electric Current & Current Densities	TB-1[4][133-135]	PPT	End Semester Exam,Home Assignment,SEM-EXAM1
9	CO1	COI-	Flipped Learning: Polarization	TB-1[5] [162-	PPT	End Semester

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
		3	and Continuity Equation	164]TB-1[5] [180-181]		Exam,SEM-EXAM1
10	CO1	COI-3	Comprehensive Analysis of Static Electric Field Problems	TB-1[5] [192-198]	PPT	ALM,End Semester Exam,SEM-EXAM1
11	CO2	COI-1	Introduction to Static Magnetic Fields	TB-1[7][261-266]	Chalk,Talk	End Semester Exam,SEM-EXAM1
12	CO2	COI-1	Biot–Savart’s Law	TB-1[7][261-266]	Chalk,Talk	End Semester Exam,SEM-EXAM1
13	CO2	COI-1	Ampere’s Circuital Law	TB-1[7] [273-279]	PPT	End Semester Exam,SEM-EXAM1
14	CO2	COI-2	Applications of Biot–Savart’s Law	TB-1[7][273-279]	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
15	CO2	COI-2	Curl of Magnetic Field & Differential Form of Ampere’s Law	TB-1[7] [273-279]	Chalk,Talk	End Semester Exam,SEM-EXAM1,Tutorial
16	CO2	COI-2	Stoke’s Theorem and Magnetic Field Relations	TB-1[7] [283-289]	PPT	End Semester Exam,SEM-EXAM1
17	CO2	COI-2	Flipped learning: Lorentz Force Equation	TB-1[8] [304-308]	Chalk,Talk	End Semester Exam,SEM-EXAM1
18	CO2	COI-3	Force on a Current Element	TB-1[10] [426- 428]	PPT	End Semester Exam,Home Assignment,SEM-EXAM1
19	CO2	COI-3	Boundary Conditions on Magnetic Fields (B and H)	TB-1[8] [304-308]	Chalk,Talk	ALM,End Semester Exam,SEM-EXAM1
20	CO2	COI-3	Comprehensive Analysis of Static Magnetic Field Problems	TB-1[10] [426- 428]	PPT	End Semester Exam,SEM-EXAM1
21	CO3	COI-1	Faraday’s Law of Electromagnetic Induction	TB-1[10] [423- 425]	PPT	End Semester Exam,SEM-EXAM2
22	CO3	COI-1	Maxwell’s Equations – Integral Form	TB-1[7][283-284]	Chalk,Talk	End Semester Exam,SEM-EXAM2
23	CO3	COI-	Maxwell’s Equations –	TB-1[7] [353-	Chalk,Talk	End Semester

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
		1	Differential Form	364]		Exam,SEM-EXAM2
24	CO3	COI-2	Wave Equation in Free Space	TB-1[10] [423- 425]	PPT	End Semester Exam,SEM-EXAM2
25	CO3	COI-2	Uniform Plane Waves – General Solution	TB-1[10] [419- 422]	PPT	ALM,End Semester Exam,SEM-EXAM2
26	CO3	COI-2	Wave Equation in Conducting Medium	TB-1[10] [417- 428]	PPT	End Semester Exam,SEM-EXAM2
27	CO3	COI-3	Wave Equations in Phasor Form	TB-1[10] [417- 428]	PPT	End Semester Exam,HA,SEM-EXAM2
28	CO3	COI-3	Flipped Learning: Wave Propagation in Lossless Medium	TB-1[10] [425- 428]	PPT	End Semester Exam,SEM-EXAM2,Tutorial
29	CO3	COI-3	Poynting Vector and Power Flow	TB-1[10] [435- 438]	PPT	End Semester Exam,SEM-EXAM2
30	CO3	COI-3	Comprehensive Analysis of EM Wave Propagation	TB-1[10] [439- 445]	PPT	ALM,End Semester Exam,SEM-EXAM2
31	CO4	COI-1	Basics of Transmission Lines	TB-5[7] [414-417]	PPT	End Semester Exam,SEM-EXAM2
32	CO4	COI-2	General Solution of Transmission Line Equations	TB-5[7] [417-420]	PPT	End Semester Exam,SEM-EXAM2
33	CO4	COI-3	Infinite Transmission Line & Wavelength	TB-5[7] [420-425]	PPT	End Semester Exam,SEM-EXAM2
34	CO4	COI-3	Reflection Coefficient	TB-5[7] [425-427]	PPT	End Semester Exam,SEM-EXAM2
35	CO4	COI-3	Flipped Learning: Open & Short-Circuited Lines	TB-5[7] [427-429]	PPT	End Semester Exam,SEM-EXAM2
36	CO4	COI-4	Lossless Transmission Line & Standing Wave Ratio	TB-5[7] [429-431]	PPT	ALM,End Semester Exam,SEM-EXAM2
37	CO4	COI-4	Input Impedance of Transmission Lines	TB-5[7] [437-443]	PPT	End Semester Exam,SEM-EXAM2,Tutorial

Sess.No.	CO	COI	Topic	Book No[CH No][Page No]	Teaching-Learning Methods	EvaluationComponents
38	CO4	COI-4	Open & Short-Circuited Lossless Lines	TB-5[7] [439-442]	PPT	End Semester Exam, Home Assignment, SEM-EXAM2
39	CO4	COI-4	Impedance Matching Techniques	TB-5[7] [437-440]	PPT	ALM, End Semester Exam, SEM-EXAM2

Lecture Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Able to understand the Course Structure and Syllabus

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
50	Introduction to Course, Course handout description, discussion on syllabus and evaluation plan	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 2 Students will be able to recall the definitions of static electric fields and identify different coordinate systems.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	PPT	--- NOT APPLICABLE ---
20	Introduction to Static Electric Fields	1	PPT	--- NOT APPLICABLE ---
20	Cartesian, Cylindrical & Spherical Coordinate Systems	1	PPT	--- NOT APPLICABLE ---
5	Session Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 3 Students will be able to define types of charge distributions and state Coulomb's law.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	Talk	--- NOT APPLICABLE ---
20	Line, Surface & Volume Charge Densities	1	Chalk	--- NOT APPLICABLE ---
20	Statement and Basic Formula of Coulomb's Law	1	PPT	--- NOT APPLICABLE ---
5	Session Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 4

Session Outcome: 4 Students will be able to recall the formulas for electric field intensity and electric flux density

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	Chalk	--- NOT APPLICABLE ---
20	Definition & Formula of Electric Field Intensity (E)	2	Talk	--- NOT APPLICABLE ---
20	Definition & Formula of Electric Flux Density (D)	2	PPT	--- NOT APPLICABLE ---
5	Session Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 5

Session Outcome: 5 Students will be able to explain Gauss's law and interpret its physical significance through simple examples.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	Chalk	--- NOT APPLICABLE ---
20	Understanding Gauss's Law	2	Chalk	One minute paper

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Applications: Infinite Line Charge, Spherical Charge Distribution	2	Talk	One minute paper
5	Session Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 6 Divergence Theorem, Electric Potential & Potential Gradient

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	PPT	--- NOT APPLICABLE ---
20	Divergence Theorem & Its EM Interpretation	2	PPT	--- NOT APPLICABLE ---
20	Electric Potential & Potential Gradient	2	PPT	--- NOT APPLICABLE ---
5	Session Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 7 Students will be able to analyze electric and flux boundary conditions at dielectric–dielectric and conductor–dielectric interfaces.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Chalk	--- NOT APPLICABLE ---
20	Normal & Tangential Boundary Conditions	3	Talk	--- NOT APPLICABLE ---
20	Dielectric–Conductor and Dielectric–Dielectric Interfaces	3	Chalk	--- NOT APPLICABLE ---
5	Session Summary	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 8 Students will be able to analyze current flow mechanisms and evaluate current density in different media.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Conduction & Convection Current	3	PPT	--- NOT APPLICABLE ---
20	Current Density and Its Vector Nature	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 9 Students will be able to derive and analyze the continuity equation for both steady and time-varying charge systems.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Charge Conservation Principle	3	PPT	--- NOT APPLICABLE ---
20	Differential and Integral Form of Continuity Equation	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 10 Students will be able to analyze field problems using Gauss's law, E-field relations, and potential gradients for complex geometries.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Solving Electric Field Problems for Non-Uniform Charge Distributions	3	PPT	Problem-Based Learning
20	Field Analysis Using Gauss's Law & Potential Method	3	PPT	Problem-Based Learning
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 11

Session Outcome: 1 Students will be able to recall the basic concepts of magnetic fields and identify magnetic field quantities.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	Chalk	--- NOT APPLICABLE ---
20	Magnetic field concepts and terminology	1	Talk	--- NOT APPLICABLE ---
20	Magnetic field vectors (B and H)	1	Chalk	--- NOT APPLICABLE ---
5	Session Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 2 Students will be able to state Biot-Savart's law and identify its fundamental expressions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	Chalk	--- NOT APPLICABLE ---
20	Biot-Savart's Law definition	1	Talk	--- NOT APPLICABLE ---
20	Magnetic field due to current elements	1	Talk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

5	Session Summary	1	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 3 Students will be able to recall Ampere's law and list conditions for its applicability.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	PPT	--- NOT APPLICABLE ---
20	Integral form of Ampere's Circuital Law	1	PPT	--- NOT APPLICABLE ---
20	Applications to symmetric current distributions	1	PPT	--- NOT APPLICABLE ---
5	Session Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 14

Session Outcome: 4 Students will be able to explain how Biot-Savart's law is applied to various current configurations

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	Talk	--- NOT APPLICABLE ---
20	Magnetic field due to a long straight conductor	2	Chalk	One minute paper
20	Field due to circular current loops	2	Talk	One minute paper
5	Session Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 15

Session Outcome: 5 Students will be able to interpret the meaning of curl and explain the differential form of Ampere's law.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	Chalk	--- NOT APPLICABLE ---
20	Curl of a vector field	2	Talk	--- NOT APPLICABLE ---
20	Differential form	2	Chalk	--- NOT APPLICABLE ---
5	Session Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 16

Session Outcome: 6 Students will be able to describe Stoke's theorem and relate integral and differential forms of Ampere's law using it.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	PPT	--- NOT APPLICABLE ---
20	Stoke's Theorem	2	PPT	--- NOT APPLICABLE ---
20	Analysis	2	PPT	--- NOT APPLICABLE ---
5	Session Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 17

Session Outcome: 7 Students will be able to analyze the force on a moving charge and relate it to magnetic field interaction.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	Chalk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

20	Lorentz force	2	Talk	--- NOT APPLICABLE ---
20	Motion of charged particles in magnetic fields	2	Chalk	--- NOT APPLICABLE ---
5	Session Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 18

Session Outcome: 8 Students will be able to analyze the magnetic force acting on various current-carrying conductors.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Force on a differential current element	3	PPT	--- NOT APPLICABLE ---
20	Force on straight and loop conductors	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 19

Session Outcome: 9 Students will be able to analyze magnetic field behavior at boundaries, including conductor–air and two-dielectric interfaces.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Chalk	--- NOT APPLICABLE ---
20	Normal and tangential boundary conditions	3	Talk	Problem-Based Learning
20	Magnetic field transition at interfaces	3	Chalk	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

5	Session Summary	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 20

Session Outcome: 10 Students will be able to apply multiple magnetic field laws to analyze complex field distributions in practical systems.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Field analysis using Biot–Savart and Ampere’s laws	3	PPT	--- NOT APPLICABLE ---
20	Multi-region magnetic field problems	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 21

Session Outcome: 1 Students will be able to recall Faraday’s law and identify its basic mathematical expressions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	PPT	--- NOT APPLICABLE ---
20	Faraday’s Law of Induction (Basic Form)	1	PPT	--- NOT APPLICABLE ---
20	EMF generation due to time-varying magnetic fields	1	PPT	--- NOT APPLICABLE ---
5	Session Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 22

Session Outcome: 2 Students will be able to state Maxwell's four equations in integral form and identify their physical meaning.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	Chalk	--- NOT APPLICABLE ---
20	Maxwell's Equations (Integral Forms)	1	Talk	--- NOT APPLICABLE ---
20	Physical interpretation of each equation	1	Chalk	--- NOT APPLICABLE ---
5	Session Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 23

Session Outcome: 3 Students will be able to recall Maxwell's equations in differential form and recognize their relation to vector calculus operations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	Chalk	--- NOT APPLICABLE ---
20	Maxwell's Equations (Differential Forms)	1	Talk	--- NOT APPLICABLE ---
20	Derivations	1	Chalk	--- NOT APPLICABLE ---
5	Session Summary	1	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 24

Session Outcome: 4 Students will be able to explain how wave equations are obtained from Maxwell's equations for free space.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	PPT	--- NOT APPLICABLE ---
20	Derivation of Wave Equation	2	PPT	--- NOT APPLICABLE ---
20	EM Wave Characteristics in Free Space	2	PPT	--- NOT APPLICABLE ---
5	Session Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 25

Session Outcome: 5 Students will be able to interpret the general solution of plane waves and explain wave propagation concepts.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	PPT	--- NOT APPLICABLE ---
20	General solution of Uniform Plane Waves	2	PPT	One minute paper
20	Propagation constant & phase velocity concepts	2	PPT	One minute paper
5	Session Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 26

Session Outcome: 6 Students will be able to describe wave behavior in conducting media and explain attenuation and skin effect.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	PPT	--- NOT APPLICABLE ---
20	Wave equation for conducting medium	2	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
20	Attenuation, skin depth, and lossy behavior	2	PPT	--- NOT APPLICABLE ---
5	Session Summary	2	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 27

Session Outcome: 7 Students will be able to analyze EM wave propagation using phasor-domain wave equations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Transforming wave equations to phasor domain	3	PPT	--- NOT APPLICABLE ---
20	Analysis of complex propagation constants	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 28

Session Outcome: 8 Students will be able to examine lossless wave conditions and analyze wave behavior when attenuation is zero.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Lossless medium conditions	3	PPT	--- NOT APPLICABLE ---
20	Phase velocity, intrinsic impedance analysis	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

SESSION NUMBER : 29

Session Outcome: 9 Students will be able to analyze power flow in EM waves using the Poynting vector and its significance.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Poynting Vector (Instantaneous & Average)	3	PPT	--- NOT APPLICABLE ---
20	Power transmission through EM fields	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 30

Session Outcome: 10 Students will be able to analyze wave propagation under various conditions using Maxwell's equations and wave equations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	EM wave propagation in free space, lossless & conducting media	3	PPT	Problem-Based Learning
20	Comparison of attenuation, reflection, and power flow	3	PPT	Problem-Based Learning
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 31

Session Outcome: 1 Students will be able to recall the fundamental concepts and identify basic parameters of transmission lines.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	1	PPT	--- NOT APPLICABLE ---
20	Transmission line fundamentals and primary constants	1	PPT	--- NOT APPLICABLE ---
20	Concept of distributed parameters	1	PPT	--- NOT APPLICABLE ---
5	Session Summary	1	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 32

Session Outcome: 2 Students will be able to state the general voltage and current solutions of the transmission line equations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	2	Talk	--- NOT APPLICABLE ---
20	General solution:	2	PPT	--- NOT APPLICABLE ---
20	Propagation constant and characteristic impedance	2	PPT	--- NOT APPLICABLE ---
5	Session Summary	2	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 33

Session Outcome: 3 Students will be able to recall the properties of an infinite line and identify wavelength and velocity of propagation.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Infinite line characteristics	3	PPT	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

20	Wavelength and wave velocity	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 34

Session Outcome: 4 Students will be able to explain reflection phenomena and interpret the reflection coefficient mathematically.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Talk	--- NOT APPLICABLE ---
20	Reflection coefficient concept	3	PPT	--- NOT APPLICABLE ---
20	Voltage & current reflections on lines	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 35

Session Outcome: 5 Students will be able to describe the behavior of open- and short-circuited lines and explain how they affect impedance.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Talk	--- NOT APPLICABLE ---
20	Input impedance of open-circuited lines	3	PPT	--- NOT APPLICABLE ---
20	Input impedance of short-circuited lines	3	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Summary	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 36

Session Outcome: 6 Students will be able to explain the concept of lossless lines and understand SWR and its significance.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	4	Talk	--- NOT APPLICABLE ---
20	Lossless line equation	4	PPT	One minute paper
20	Standing Wave Ratio	4	PPT	One minute paper
5	Session Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 37

Session Outcome: 7 Students will be able to analyze the input impedance for various line conditions and lengths.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	4	Talk	--- NOT APPLICABLE ---
20	Input impedance expression for any line length	4	PPT	--- NOT APPLICABLE ---
20	Practical evaluation of input impedance	4	PPT	--- NOT APPLICABLE ---
5	Session Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 38

Session Outcome: 8 Students will be able to analyze impedance behavior of lossless lines under open and short conditions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	4	Talk	--- NOT APPLICABLE ---
20	Analytical behavior of Z_{in} for open lossless line	4	PPT	--- NOT APPLICABLE ---
20	Analytical behavior of Z_{in} for short lossless line	4	PPT	--- NOT APPLICABLE ---
5	Session Summary	4	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 39

Session Outcome: 9 Students will be able to analyze various impedance matching methods used in RF and communication systems.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	4	PPT	--- NOT APPLICABLE ---
20	Quarter wave transformer matching	4	PPT	Problem-Based Learning
20	Stub matching (single and double stub)	4	PPT	Problem-Based Learning
5	Session Summary	4	PPT	--- NOT APPLICABLE ---

Tutorial Course DELIVERY Plan:

List of Experiments supposed to finish in Open Lab Sessions:

Lab session no	List of Experiments	CO-Mapping
1	Applying Gauss's Law & Divergence Concepts	CO1
2	Applying Electric Potential & Potential Gradient	CO1

Lab session no	List of Experiments	CO-Mapping
3	Applying Boundary Conditions & Continuity Equation	CO1
4	Applying Biot–Savart’s Law & Ampere’s Law	CO2
5	Applying Curl & Differential Form of Ampere’s Law	CO2
6	Applying Lorentz Force, Force on Current Element & Boundary Conditions	CO2
7	Applying Faraday’s Law & Maxwell’s Equations	CO3
8	Applying Wave Equation & Uniform Plane Wave Concepts	CO3
9	Applying Phasor Wave Equations, Lossless Propagation & Poynting Vector	CO3
10	Applying General Solution & Input Impedance Calculations	CO4
11	Applying Reflection Coefficient & SWR Concepts	CO4
12	Analyzing Open/Short Lines & Lossless Line Behavior	CO4
13	Analyzing Impedance Matching Techniques	CO4

Tutorial Session wise Teaching – Learning Plan

SESSION NUMBER : 1

Session Outcome: 1 Students will be able to apply Gauss’s law and divergence to compute electric fields and charge densities for symmetric charge distributions.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Chalk	--- NOT APPLICABLE ---
20	Application of Gauss’s Law to line, surface & volume charge distributions	3	Talk	--- NOT APPLICABLE ---
20	Using Divergence to obtain charge density	3	Chalk	--- NOT APPLICABLE ---
5	Session Summary	3	Talk	--- NOT APPLICABLE ---

SESSION NUMBER : 2

Session Outcome: 2 Students will be able to calculate electric potential for different charge configurations and apply potential gradient to find electric field intensity.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Talk	--- NOT APPLICABLE ---
20	Computing Electric Potential for point, line & sheet charges	3	Chalk	--- NOT APPLICABLE ---
20	Applying Potential Gradient ($E = -\nabla V$) in Cartesian, cylindrical & spherical coordinates	3	Talk	--- NOT APPLICABLE ---
5	Session Summary	3	Chalk	--- NOT APPLICABLE ---

SESSION NUMBER : 3

Session Outcome: 3 Students will be able to apply boundary conditions to determine field components at interfaces and use continuity equation to analyze current flow.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Talk	Problem-Based Learning
20	Applying Boundary Conditions on E and D at dielectric–dielectric and conductor–dielectric interfaces	3	Chalk	Problem-Based Learning
20	Applying the Continuity Equation to current density problems	3	Chalk	Problem-Based Learning
5	Session Summary	3	Talk	Problem-Based Learning

SESSION NUMBER : 4

Session Outcome: 1 Students will be able to apply Biot–Savart’s law and Ampere’s circuital law to compute magnetic fields for various current configurations.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	Talk	Problem-Based Learning
20	Application of Biot–Savart’s Law to long straight wire, circular loop, and finite-length conductor	3	Chalk	Problem-Based Learning
20	Applying Ampere’s Circuital Law to symmetrical structures (solenoids, toroids, coaxial cables)	3	Talk	Problem-Based Learning
5	Session Summary	3	Chalk	Problem-Based Learning

SESSION NUMBER : 5

Session Outcome: 2 Students will be able to apply curl operations and use the differential form of Ampere’s law to determine current densities and field variation.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Evaluating Curl of Magnetic Field ($\nabla \times \mathbf{H}$) for given vector fields	3	PPT	--- NOT APPLICABLE ---
20	Using Differential Ampere’s Law ($\nabla \times \mathbf{H} = \mathbf{J}$) to compute current density in conductors	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 6

Session Outcome: 3 Students will be able to apply Lorentz force concepts and boundary conditions to analyze magnetic force and magnetic field behavior across interfaces.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Calculating Forces: Lorentz force on moving charge & force on a differential/current-carrying element	3	PPT	--- NOT APPLICABLE

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

20	Applying Boundary Conditions on B and H at conductor–air and magnetic material interfaces	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 7

Session Outcome: 1 Students will be able to apply Faraday’s law and Maxwell’s equations to solve problems involving time-varying electric and magnetic fields.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Applying Faraday’s Law to compute induced EMF in coils and loops	3	PPT	Problem-Based Learning
20	Using Maxwell’s Equations (integral and differential forms) to determine field relationships	3	PPT	Problem-Based Learning
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 8

Session Outcome: 2 Students will be able to apply EM wave equations and uniform plane wave relations to evaluate propagation in free space and conducting media.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Applying Wave Equation in Free Space & Conducting Medium to find field components	3	PPT	--- NOT APPLICABLE ---
20	Applying General Solution of Uniform Plane Waves to compute propagation constant, wavelength, and velocity	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 9

Session Outcome: 3 Students will be able to apply phasor-domain wave equations, analyze lossless propagation, and compute power flow using the Poynting vector.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Using Phasor Form of Wave Equations to compute amplitude and phase variations	3	PPT	--- NOT APPLICABLE ---
20	Applying Poynting Vector to evaluate instantaneous and average power in EM waves	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 10

Session Outcome: 1 Students will be able to apply the general solution of transmission lines and compute input impedance for different line lengths.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	Problem-Based Learning
20	Applying the general solution for the given loads	3	PPT	Problem-Based Learning
20	Computing input impedance	3	PPT	Problem-Based Learning
5	Session Summary	3	PPT	Problem-Based Learning

SESSION NUMBER : 11

Session Outcome: 2 Students will be able to apply reflection coefficient and SWR formulas to determine voltage standing wave patterns.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	3	PPT	--- NOT APPLICABLE ---
20	Calculating reflection coefficient	3	PPT	--- NOT APPLICABLE ---
20	Applying SWR relation	3	PPT	--- NOT APPLICABLE ---
5	Session Summary	3	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 12

Session Outcome: 3 Students will be able to analyze input impedance of open and short-circuited lines and evaluate how lossless properties affect standing waves.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	4	PPT	--- NOT APPLICABLE ---
20	Analyzing Z_{in} of open & short-circuited lossless lines	4	PPT	--- NOT APPLICABLE ---
20	Evaluating standing wave patterns and their frequency dependence	4	PPT	--- NOT APPLICABLE ---
5	Session Summary	4	PPT	--- NOT APPLICABLE ---

SESSION NUMBER : 13

Session Outcome: 4 Students will be able to analyze impedance matching problems using quarter-wave transformers and single/double stub matching.

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods
5	Session Introduction	4	PPT	--- NOT APPLICABLE ---
20	Quarter-wave transformer analysis	4	PPT	--- NOT APPLICABLE ---

Time(min)	Topic	BTL	Teaching-Learning Methods	Active Learning Methods

20	Analyzing single-stub and double-stub matching for various loads	4	PPT	--- NOT APPLICABLE ---
5	Session Summary	4	PPT	--- NOT APPLICABLE ---

Practical Course DELIVERY Plan:

NO Delivery Plan Exists

Practical Session wise Teaching – Learning Plan

No Session Plans Exists

Skilling Course DELIVERY Plan:

NO Delivery Plan Exists

Skilling Session wise Teaching – Learning Plan

No Session Plans Exists

WEEKLY HOMEWORK ASSIGNMENTS/ PROBLEM SETS/OPEN ENDED PROBLEM-SOLVING EXERCISES etc:

Week	Assignment Type	Assignment No	Topic	Details	co
1	Weekly Homework Assignments	1	Coordinate systems, Charge distributions	1. Explain how electric field intensity varies for point, line, and surface charge distributions. Discuss the nature of field lines, dependence on distance, and the physical meaning of intensity in each case. 2. Describe the concept of electric flux density (D) and explain its relationship with electric field (E) in different media. Include examples showing how permittivity affects the magnitude of D.	CO1
2	Weekly Homework Assignments	2	Electric current, current density & continuity equation, Boundary conditions of E and D	1. Explain Gauss's Law and illustrate how it relates to the divergence of the electric	CO1

Week	Assignment Type	Assignment No	Topic	Details	co
				flux density. Discuss its physical significance and give an example of a symmetric charge distribution. 2. Describe the boundary conditions on electric field (E) and electric flux density (D) at the interface of two dielectric media. Explain the significance of normal and tangential components and relate them to physical charge behavior at interfaces.	
3	Weekly Homework Assignments	3	Biot–Savart’s Law & Ampere’s Circuital Law Differential Form of Ampere’s Law, Curl & Stokes’ Theorem	1.Explain how Biot–Savart’s Law is used to determine the magnetic field due to: a) A finite-length straight conductor b) A circular conducting loop Discuss the physical meaning of each term in the equation and describe why this law is essential for computing magnetic fields in current-carrying systems. 2.Differentiate between the integral and differential forms of Ampere’s Circuital Law. Explain how Stokes’ Theorem is used to convert one form into the other, and interpret the physical meaning of the curl of magnetic field intensity (H).	CO2
4	Weekly Homework Assignments	4	Lorentz Force Equation & Force on Current Element Boundary Conditions on Magnetic Fields (B and H)	1. Describe the Lorentz force equation and explain how it is applied to determine the force experienced by: a) A moving charge in a magnetic field b) A current-carrying conductor placed in a uniform magnetic field Explain the significance of directions using the right-hand rule. 2. Explain the boundary conditions on magnetic flux density (B) and magnetic field intensity	CO2

Week	Assignment Type	Assignment No	Topic	Details	co
				(H) at the interface between two magnetic media. Discuss how tangential and normal components behave when transitioning across boundaries with different permeabilities.	
6	Weekly Homework Assignments	5	Faraday's Law of Induction Maxwell's Equations (Integral & Differential Forms), Wave Equation for Free Space	1. Using Faraday's law, determine: a) The induced EMF in the loop b) The induced current	CO3
8	Weekly Homework Assignments	6	Uniform Plane Wave Propagation Wave Propagation in Conducting and Lossless Media, Poynting Vector	1. For an uniform plane wave calculate phase constant, attenuation constant and skin depth	CO3
10	Weekly Homework Assignments	7	Transmission lines	Calculate the input impedance and characteristic impedance of an infinite length transmission line	CO4
12	Weekly Homework Assignments	8	Impedance matching	Problems of stub matching (single and double stubs)	CO4

COURSE TIME TABLE:

	Hour	1	2	3	4	5	6	7	8	9
Day	Component									
Mon	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Tue	Theory	-- -	-- -	V-S51,V-S52,V-S53,V-S54	V-S51,V-S52	---	---	-- -	-- -	-- -
	Tutorial	-- -	-- -	--	V-S53,V-S54	---	---	-- -	-- -	-- -
	Lab	-- -	-- -	--	--	---	---	-- -	-- -	-- -
	Skilling	-- -	-- -	--	--	---	---	-- -	-- -	-- -
Wed	Theory	-- -	-- -	---	---	V-S1,V-S2,V-S3,V-S4	V-S1,V-S2	-- -	-- -	-- -
	Tutorial	-- -	-- -	---	---	--	V-S3,V-S4	-- -	-- -	-- -
	Lab	-- -	-- -	---	---	--	--	-- -	-- -	-- -

	Hour	1	2	3	4	5	6	7	8	9
	Skilling	-- -	-- -	---	---	--	--	-- -	-- -	-- -
Thu	Theory	-- -	-- -	---	---	V-S51,V-S52,V-S53,V-S54	V-S53,V-S54	-- -	-- -	-- -
	Tutorial	-- -	-- -	---	---	--	V-S51,V-S52	-- -	-- -	-- -
	Lab	-- -	-- -	---	---	--	--	-- -	-- -	-- -
	Skilling	-- -	-- -	---	---	--	--	-- -	-- -	-- -
Fri	Theory	-- -	-- -	---	---	V-S1,V-S2,V-S3,V-S4	V-S3,V-S4	-- -	-- -	-- -
	Tutorial	-- -	-- -	---	---	--	V-S1,V-S2	-- -	-- -	-- -
	Lab	-- -	-- -	---	---	--	--	-- -	-- -	-- -
	Skilling	-- -	-- -	---	---	--	--	-- -	-- -	-- -
Sat	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--
Sun	Theory	--	--	--	--	--	--	--	--	--
	Tutorial	--	--	--	--	--	--	--	--	--
	Lab	--	--	--	--	--	--	--	--	--
	Skilling	--	--	--	--	--	--	--	--	--

REMEDIAL CLASSES:

Supplement course handout, which may perhaps include special lectures and discussions that would be planned, and schedule notified according

SELF-LEARNING:

Assignments to promote self-learning, survey of contents from multiple sources.

S.no	Topics	CO	ALM	References/MOOCs
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DELIVERY DETAILS OF CONTENT BEYOND SYLLABUS:

Content beyond syllabus covered (if any) should be delivered to all students that would be planned, and schedule notified accordingly.

S.no	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
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EVALUATION PLAN:

Evaluation Type	Evaluation Component	Weightage/Marks		Assessment Dates	Duration (Hours)	CO1	CO2	CO3	CO4
End Semester Summative Evaluation Total= 40 %	End Semester Exam	Weightage	40		180	10	10	10	10
		Max Marks	100			25	25	25	25
In Semester Formative Evaluation Total= 26 %	Home Assignment and Textbook	Weightage	5		60	1.25	1.25	1.25	1.25
		Max Marks	100			25	25	25	25
	Tutorial	Weightage	15		60	3.75	3.75	3.75	3.75
		Max Marks	100			25	25	25	25
	ALM	Weightage	6		60	1.5	1.5	1.5	1.5
		Max Marks	100			25	25	25	25
In Semester Summative Evaluation Total= 34 %	Semester in Exam-II	Weightage	17		90			8.5	8.5
		Max Marks	50					25	25
	Semester in Exam-I	Weightage	17		90	8.5	8.5		
		Max Marks	50			25	25		

ATTENDANCE POLICY:

Every student is expected to be responsible for regularity of his/her attendance in class rooms and laboratories, to appear in scheduled tests and examinations and fulfill all other tasks assigned to him/her in every course. In every course, student has to maintain a minimum of 85% attendance to be eligible for appearing in Semester end examination of the course, for cases of medical issues and other unavoidable circumstances the students will be condoned if their attendance is between 75% to 85% in every course, subjected to submission of medical certificates, medical case file and other needful documental proof to the concerned departments.

DETENTION POLICY :

In any course, a student has to maintain a minimum of 85% attendance and In-Semester Examinations to be eligible for appearing to the Semester End Examination, failing to fulfill these conditions will deem such student to have been detained in that course.

PLAGIARISM POLICY :

Supplement course handout, which may perhaps include special lectures and discussions

COURSE TEAM MEMBERS, CHAMBER CONSULTATION HOURS AND CHAMBER VENUE DETAILS:

Supplement course handout, which may perhaps include special lectures and discussions

Name of Faculty	Delivery Component of Faculty	Sections of Faculty	Chamber Consultation Day (s)	Chamber Consultation Timings for each day	Chamber Consultation Room No:	Signature of Course faculty:
Venkata Narayana Madhavareddy	L	1-MA	-	-	-	-
Venkata Narayana Madhavareddy	T	1-MA	-	-	-	-
Zia UR Rahman Mohammad	L	3-MA,51-MA	-	-	-	-
Zia UR Rahman Mohammad	T	3-MA,51-MA	-	-	-	-
Chinnari Sri Kavya Korada	L	53-MA	-	-	-	-
Chinnari Sri Kavya Korada	T	53-MA	-	-	-	-
Sujatha Moorthy	L	4-MA,52-MA	-	-	-	-
Sujatha Moorthy	T	4-MA,52-MA	-	-	-	-
VENKATA KISHORE KOTHAPUDI	L	2-MA,54-MA	-	-	-	-
VENKATA KISHORE KOTHAPUDI	T	2-MA,54-MA	-	-	-	-

GENERAL INSTRUCTIONS

Students should come prepared for classes and carry the text book(s) or material(s) as prescribed by the Course Faculty to the class.

NOTICES

Most of the notices are available on the LMS platform.

All notices will be communicated through the institution email.

All notices concerning the course will be displayed on the respective Notice Boards.

Signature of COURSE COORDINATOR

(Zia UR Rahman Mohammad)(3413)(ECE)

Signature of Department Prof. Incharge Academics & Vetting Team Member

Department Of ECE

HEAD OF DEPARTMENT:

Approval from: DEAN-ACADEMICS

(Sign with Office Seal)

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